THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · NCERT-TYPE UNIT TEST SET 3

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Two triangles have angles $50^\circ, 60^\circ, 70^\circ$ each. They are: 1

(a) congruent by SSS **(b)** never congruent
(c) not necessarily congruent **(d)** always congruent

2. How many pieces of length $\frac{1}{2}$ m can be cut from a rope of 7 m? 1

(a) 14 **(b)** 28 **(c)** $3\frac{1}{2}$ **(d)** 7

3. In a linear pair one angle is 4 times the other. The smaller angle is: 1

(a) 36° **(b)** 46° **(c)** 26° **(d)** 56°

4. A two-digit number has digits 3 and 6. The sum of the number and its reverse is: 1

(a) 101 **(b)** 99 **(c)** 100 **(d)** 98

5. Two sides of a triangle are 6 cm and 12 cm. The third side could be: 1

(a) 18 cm **(b)** 20 cm **(c)** 10 cm **(d)** 6 cm

6. Scaling a drawing by $\frac{2}{3}$: a line of 21 cm becomes (cm): 1

(a) 28 **(b)** 14 **(c)** $31\frac{1}{2}$ **(d)** 7

7. Two parallel lines are cut by a transversal. An interior angle is 80°. The interior angle on the same side is: 1

(a) 110° **(b)** 100° **(c)** 120° **(d)** 90°

8. A two-digit number has digits 7 (tens) and 3 (units). Number minus its reverse is: 1

(a) 37 **(b)** 38 **(c)** 36 **(d)** 35

9. The angles of a triangle are x°, $2x^\circ$ and $6x^\circ$. The value of x is: 1

(a) 21 **(b)** 20 **(c)** 22 **(d)** 19

10. The number of $\frac{1}{3}$s in $\frac{2}{3}$ is: 1

(a) 1 **(b)** 2 **(c)** 3 **(d)** $\frac{2}{9}$

11. If a transversal makes interior angles on the same side that add up to 180°, the lines are: 1

(a) perpendicular **(b)** unequal **(c)** intersecting **(d)** parallel

12. The number of steps the Collatz sequence from 6 takes to reach 1 is: 1

(a) 9 **(b)** 7 **(c)** 10 **(d)** 8

13. The exterior angle of a triangle is $(3x + 20)^\circ$ and the interior opposite angles are 80° and 45°. The value of x is: 1

(a) 35 **(b)** 34 **(c)** 37 **(d)** 36

14. $2\frac{1}{3} \times 6$ equals: 1

(a) 7 **(b)** $12\frac{1}{3}$ **(c)** 28 **(d)** 14

15. Angles A and B are vertically opposite and $A = 64^\circ$. Then $B =$ 1

(a) 74° **(b)** 54° **(c)** 64° **(d)** 84°

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. In $\triangle ABC$, side BC is extended to D, $\angle ACD = (5x - 10)^\circ$, $\angle A = (2x + 20)^\circ$ and $\angle B = (x + 30)^\circ$. Find x and $\angle ACD$. 2

17. What fraction multiplied by $\frac{2}{5}$ gives $\frac{4}{15}$? 2

18. Two supplementary angles are such that one is 3 times the other. Find them. 2

19. Show that $(10a + b) + (10b + a) = 11(a + b)$ and verify for $a = 4$, $b = 7$. 2

20. Two sides of a triangle are 4 cm and 13 cm. Find the least and the greatest whole-number length of the third side. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. In isosceles $\triangle ABC$ with $AB = AC$, $\angle A = (x + 20)^\circ$ and $\angle B = (2x - 15)^\circ$. Find x and all three angles. 3

22. A rectangle has sides $\frac{5}{6}$ m and $\frac{3}{10}$ m. Show its area as the product of the sides and compare it with the area of the smaller side squared. 3

23. A transversal cuts two lines and makes alternate interior angles of 70° and 75°. Are the lines parallel? Give a reason. 3

24. Find how many reverse-and-add steps turn 49 into a palindrome, showing each step. 3

OR

In $\triangle ABC$ and $\triangle PQR$, $AB = PQ = 6$ cm, $BC = QR = 9$ cm and $\angle B = \angle Q = 40^\circ$. Name the congruence rule, and find PR if $AC = 12$ cm. Find also the perimeter of $\triangle PQR$. 3

25. Divide $1\frac{2}{3}$ by $\frac{5}{6}$ and write the answer as a mixed number. 3

OR

Two complementary angles are in the ratio $1 : 5$. Find them. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A transversal cuts two parallel lines and one of the eight angles is 110°. How many of the eight angles measure 110° and how many measure 70°? Give reasons. 5

OR

For the three-digit number 945, find its reverse, the difference, and show that the difference is divisible by 99. 5

27. The angles of a triangle are $(x + 20)^\circ$, $(x - 10)^\circ$ and $(2x + 10)^\circ$. Find x, the three angles and classify the triangle. Also find the three exterior angles (one at each vertex). 5

OR

A journey of 90 km is done in three parts. The first part is $\frac{1}{3}$ of the journey; $\frac{3}{4}$ of the remaining distance is covered next. Find both distances and the distance still left. 5

END OF QUESTION PAPER **Total: 50 Marks**

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